

This Time Is Different?

Explainable Deep Learning of Crisis Transmission into Irish Stock Market Volatility: The Global Financial Crisis, the Irish Sovereign Debt Crisis and COVID-19

Prepared for Shahram · Version 1.4 · 30 September 2026 (status and results added; Amendment A1: two exploratory extensions; corrections) · first version 24 September 2026

Academic integrity: This is a planning document: design decisions, checks, reading and learning plan. It is not dissertation text. Write every chapter in your own words, cite every source you use, and declare AI assistance exactly as DBS policy requires (see Section 11).

1. At a glance

Item	Final decision
Title	This Time Is Different? Explainable Deep Learning of Crisis Transmission into Irish Stock Market Volatility: The Global Financial Crisis, the Irish Sovereign Debt Crisis and COVID-19
Main question	Can an explainable LSTM forecast Irish stock market volatility better than GARCH and HAR, and do its explanations reveal how historically different crises were transmitted to Ireland?
Target	Forward 5-day realised variance of the ISEQ Overall index, from close-to-close returns
Inputs	Three markets = three channels: domestic (ISEQ), United States (S&P 500), euro area (DAX). FTSE 100 replaces DAX in one robustness run only
Sample	January 2003 – August 2023 main sample (5,243 complete trading days); Sept 2023 – Dec 2025 robustness only
Crises compared	GFC (US credit) · Irish sovereign debt crisis (sovereign–bank loop) · COVID-19 (exogenous) + validation: the 2022 war/energy shock (euro channel); Brexit 2016 (UK) in the FTSE robustness run
Models	GARCH(1,1), HAR, LSTM (PyTorch)
Tests	MSE + QLIKE losses; Diebold–Mariano with Harvey–Leybourne–Newbold correction
Explainability	GradientSHAP (Captum), attributions grouped by transmission channel, stability across 5 seeds
Original contribution	(1) SHAP ‘fingerprints’ compared across crises of different origin as falsifiable hypotheses; (2) a ‘learning from history’ test of whether models trained on past crises survive the next one; (3) applied to a small open economy (Ireland)
Deferred to PhD	Monte Carlo uncertainty (MC Dropout vs GARCH simulation) · time-varying (TVP-VAR) connectedness · Irish bank stocks · Irish sovereign bonds
Exploratory extensions (added 30 Sep 2026)	E1 Diebold–Yilmaz spillovers vs SHAP; E2 volatility paradox (Danielsson, Valenzuela & Zer, 2018). Written into PREREGISTRATION.md as Amendment A1 and committed before running; labelled exploratory
Status	All core steps run (benchmarks, LSTM, evaluation, SHAP). Robustness: Parkinson done; FTSE, look-back 66 and post-2023 still to run. Exploratory E1–E2 done (30 Sep 2026). Results: Section 12; full tables in the README and the Project Handbook.

2. Final title and why

Final title: This Time Is Different? Explainable Deep Learning of Crisis Transmission into Irish Stock Market Volatility: The Global Financial Crisis, the Irish Sovereign Debt Crisis and COVID-19

- **Main title** — borrows the title of Reinhart & Rogoff (2009), the best-known history of financial crises, as a question. It signals a historical question answered with quantitative methods, which is exactly the contribution (research question 3 tests it directly).

- **‘Explainable Deep Learning’** — names the method (LSTM + SHAP) and puts explainability at the centre, in the language of explainable-AI research.
- **‘Crisis Transmission’** — the contagion/spillover language of the financial-econometrics literature.
- **‘Irish Stock Market Volatility’** — the Irish angle and the volatility-forecasting field.
- **‘The Global Financial Crisis, the Irish Sovereign Debt Crisis and COVID-19’** — names the three crises. Putting Ireland’s own crisis in the title makes the Irish angle impossible to miss, and all three fall in the reliable part of the data.

Titles considered and rejected:

Alternative	Why rejected
...from the Dot-com Crash to COVID-19	Dot-com dropped as an evaluation window (weak Irish signal, too little earlier data).
...Uncertainty-Aware, Explainable...	Monte Carlo uncertainty is deferred to the PhD; do not promise it in the title.
Explainability as a Diagnostic for Crisis Transmission...	Strong, but loses the historical hook; keep as a sentence in the Introduction instead.
...Irish Equity Volatility...	Fine; ‘Stock Market’ is clearer to a general reader. Either is acceptable.

3. Verification log — every claim checked

Status key: VERIFIED = confirmed by data or a primary source in this session · CORRECTED = an earlier claim was wrong and has been fixed · RISK = real problem, mitigation decided · DEFERRED = moved to PhD · CONFIRM = believed correct, you must confirm before citing · NEW = found during checking.

3A. Data

#	Claim	Status	Evidence	Action
1	Yahoo has daily ISEQ Overall (^ISEQ) back to the 1990s	VERIFIED	Data exist from 1997; the study uses Jan 2003 – Aug 2023: 5,243 complete days	Use as target
2	Feature series cover the sample	VERIFIED	^GSPC, ^GDAXI (and ^FTSE for robustness) available from 1997; downloaded from Oct 2002	Use
3	Euro Stoxx 50 can represent the euro area	CORRECTED	^STOXX50E on Yahoo only from 2007	Use DAX (^GDAXI) as euro-area core
4	VSTOXX (European fear index) available	CORRECTED	^V2TX: ‘Not Found’ on Yahoo	No fear index used
5	Garman–Klass volatility from ISEQ OHLC is reliable	CORRECTED	Yahoo ISEQ ‘open’ equals previous close on 37.7% of days (2007–12) and 75.4% (2013–19); monthly corr(GK, r^2) only 0.54 in 2003–06	Rejected. Target = forward 5-day realised variance from closes
6	Parkinson (high–low) estimator usable	VERIFIED	Monthly corr with r^2 : 0.98–0.99 after 2007, 0.67 in 2003–06	Robustness check, 2007+ only
7	Foreign data can be aligned without look-ahead	VERIFIED	As-of join on date $\leq t$; values carried forward on only 0.9–2.7% of Irish days	Forecast issued before Dublin open on $t+1$
8	Dot-com data should be used	CORRECTED	Only 668 days (2.7 years) before the Nasdaq peak; Irish median vol 14.9% vs 13.1% calm. A 2005 start would leave only 659 days before the GFC — the same problem	Sample starts Jan 2003, after the dot-com trough: 1,166 days (4.6 years) before the GFC
9	ISEQ is a consistent index through 2025	RISK	CRH left Sept 2023, Flutter Jan 2024, Smurfit Kappa Jul 2024; the trio were ~€74bn of the ~€152bn ISEQ 20 (Irish Times)	Main sample ends 31 Aug 2023; 592 later days = robustness
10	Zero returns will not break log models	VERIFIED	Only 2 zero-return days in the sample; target never zero	Floor daily r^2 at $1e-8$ before logs
11	The crises are visible in Irish data	VERIFIED	Max drawdowns: GFC –78.0%, Irish debt crisis –32.3%, COVID –39.8%, 2022 –27.5%, Brexit –18.2%; calm median vol 13.0%	Report in Chapter 4
12	Single-origin shocks visible	NEW	Brexit: vol peaks at 97.7% (June 2016). 2022 war/rate shock: median 23.3%, peak 66.5%	Two validation events with one expected channel each

13	Free daily Irish sovereign bond yields	DEFERRED	FRED has monthly only	PhD extension
14	Irish bank stocks available	CONFIRM	BIRG.IR from July 2001. A5G.IR shows data from 2000 but AIB re-listed in 2017 — check history before use	Deferred (stretch)
31	A long-run Irish share-price series exists for the volatility-paradox check	VERIFIED	FRED/OECD SPASTT01IEM661N, monthly from Jan 1955; monthly averages of daily closes — monthly-return correlation 0.9996 with the Yahoo ISEQ (2002–2025). US from 1957, Germany from 1960	Used only for exploratory E2

3B. Methods

#	Claim	Status	Evidence	Action
15	LSTM can beat econometric volatility models	VERIFIED	Bucci (2020, JFEC 18(3)): on monthly S&P 500 data (1950–2017), LSTM and NARX beat other networks and econometric models in MSE and QLIKE, also in the 2007–09 crisis. Christensen, Siggaard & Veliyev (2023, JFEC 21(5)): ML competitive with and beating HAR	Expect modest gains; a negative result is still valid
16	GARCH(1,1) is a weak benchmark	CORRECTED	Hansen & Lunde (2005): GARCH(1,1) is very hard to beat	Treat GARCH and HAR as serious benchmarks
17	SHAP works easily on a Keras/TensorFlow LSTM	RISK	Several open GitHub issues on shap DeepExplainer/GradientExplainer with TF2 LSTMs	Build the LSTM in PyTorch; use Captum GradientShap (expected-gradients SHAP). Toy test in week 2
18	KernelSHAP over the whole sample is feasible	CORRECTED	Far too slow for sequence inputs	Gradient-based SHAP on a stratified sample of days per window
19	Summing SHAP values by group is valid	VERIFIED	Additivity: attributions sum to the prediction minus baseline, so group sums are a valid decomposition; they are not a ‘group Shapley value’	Say this precisely in Methodology
20	Correlated inputs distort SHAP	VERIFIED	Aas, Jullum & Løland (2021); Kumar et al. (2020). In your data: Nasdaq–S&P 0.95, VIX–S&P –0.73, FTSE–DAX 0.84	Keep only three markets: ISEQ, S&P 500, DAX; FTSE only as a swap-in robustness check
21	Squared-error loss is enough for volatility	CORRECTED	Patton (2011, J. Econometrics 160(1)): only some losses rank forecasts correctly with a noisy proxy	Report MSE and QLIKE; QLIKE is primary
22	Plain Diebold–Mariano works for 5-day forecasts	CORRECTED	Overlapping multi-step forecasts are autocorrelated	HAC variance with 4 lags + Harvey–Leybourne–Newbold correction
23	Daily LSTM retraining is feasible	CORRECTED	Around 5,000 retrains would never finish	Expanding window, retrain once a year (~17 fits × 5 seeds)
24	Diebold–Yilmaz is a quick library call	DONE (E1)	It needs the generalised variance decomposition (Pesaran–Shin); statsmodels provides the orthogonalised one	Implemented 30 Sep 2026 as exploratory E1 with my own generalized-FEVD code, checked against statsmodels on simulated data (coefficients, MA matrices and Cholesky FEVD match; order-invariant)
25	Crisis labels could be inputs	CORRECTED	Dated after the event = look-ahead	Labels used only to split evaluation, never as features
32	Diebold & Yilmaz (2012) set-up	VERIFIED	Full text (IJF 28(1), 57–66): daily range-based volatility of four US asset classes, 25 Jan 1999 – 29 Jan 2010; VAR(4); generalized FEVD; 10-day horizon; 200-day rolling windows	E1 follows it, but with log 5-day realised variance (stale ISEQ opens)
33	Danielsson, Valenzuela & Zer (2018) method	VERIFIED	Full text (RFS 31(7), accepted version): annual volatility from 12 monthly returns July–June; one-sided HP trend, $\lambda = 5,000$; below-trend part averaged over 5 years; 60 countries, up to 211 years	E2 follows it market by market (no panel logit)

3C. Literature and novelty

#	Claim	Status	Evidence	Action
26	Twomey's Irish contagion paper is from 2018	CORRECTED	Corbet, S. & Twomey, C. (2015) 'European Equity Market Contagion: An Empirical Application to Ireland's Sovereign Debt Crisis', European Financial and Accounting Journal, 2015(3), 15–34. DCC-GARCH; strong contagion 2007–10 that eased after the troika programme	Cite correctly; compare your euro-crisis results with it
27	Twomey published on UK→Irish volatility spillovers	CORRECTED	Gallagher, L. & Twomey, C. (1998) 'Identifying the source of mean and volatility spillovers in Irish equities: a multivariate GARCH analysis', The Economic and Social Review, 29(4), 341–356 (found by the 28 Sep search; checked 30 Sep)	Cite as Gallagher & Twomey (1998)
28	Nobody has combined LSTM, SHAP and crisis volatility	CORRECTED	Sahiner (2023, Computational Economics): volatility spillovers across major crashes 1997–2021 with DCC, BEKK, Diebold–Yilmaz and a deep-learning early-warning model. Ahmed (2026, SSRN preprint): LSTM-Attention-SHAP volatility framework	Narrow the novelty claim (see 5.12); cite both as closest work

4. What changed from the earlier plan

- **Target:** Garman–Klass → forward 5-day realised variance (Yahoo's ISEQ opening prices are unreliable).
- **Sample end:** 2025 → 31 August 2023 (ISEQ lost CRH, Flutter and Smurfit).
- **Software:** Keras → PyTorch + Captum (SHAP compatibility with LSTMs).
- **Evaluation:** RMSE only → MSE + QLIKE; Diebold–Mariano with the HLN correction.
- **Scope:** Diebold–Yilmaz and Monte Carlo intervals deferred to the PhD; SHAP stability kept.
- **Crises (v1.1):** three core crises (GFC, Irish sovereign debt, COVID) plus two validation events (Brexit 2016, 2022 war/rate shock).
- **Sample (v1.2):** starts January 2003; all dot-com data removed. Not 2005, because a 2005 start leaves only 659 days before the GFC — too little to train an LSTM.
- **Markets (v1.3):** seven series → three (ISEQ, S&P 500, DAX). Nasdaq and VIX duplicate the S&P 500; the US 10-year yield barely relates to the ISEQ (0.05); FTSE overlaps DAX (0.84), so it is used only in a robustness run. Cleaner SHAP channels; same research questions.
- **New:** the 'learning from history' question made central.
- **Literature:** Twomey paper corrected to Corbet & Twomey (2015); closest prior work (Sahiner 2023; Ahmed 2026) identified; novelty claim narrowed.
- **v1.4 (30 Sep 2026):** results are in (Section 12). Two exploratory extensions added through Amendment A1: Diebold–Yilmaz moves from 'deferred to PhD' to exploratory E1 (compared with SHAP), and a new E2 checks the volatility paradox with FRED data from 1955.
- **v1.4 corrections:** Gallagher & Twomey (1998), not Twomey (1998); the tuned LSTM is look-back 22, 64 units, 2 layers; SHAP explains every day in each window plus every 10th calm day (not a random sample).

5. Final research design

5.1 Research questions and pre-registered hypotheses

Write these into a file in your project (PREREGISTRATION.md) and commit it to Git before running any model. The commit date proves the hypotheses came before the results, which examiners trust.

RQ	Question	Hypothesis (stated before results)	Test
RQ1 Accuracy	Does the LSTM forecast Irish volatility better than GARCH(1,1) and HAR, overall and within each crisis?	H1: LSTM has lower QLIKE than both benchmarks out-of-sample. H0: no difference.	Diebold–Mariano–HLN, 5% level; per-window results reported descriptively (small samples)
RQ2 Fingerprints	Do grouped SHAP attributions shift across crises in line with each crisis's origin?	Relative to calm periods: GFC → the US share rises most (domestic also high, as Ireland's banking collapse was partly home-grown); Irish debt crisis → euro and domestic shares rise, US share below its GFC level; COVID → all three channels rise together. Validation: 2022 → euro share rises; FTSE robustness run: Brexit → UK share rises.	Compare group shares by window with block-bootstrap confidence intervals
RQ3 History	Does a model trained only on earlier history keep working when a new crisis arrives?	A Minsky test. The GFC model, trained only on the calm 2003–07 boom, loses the most accuracy relative to HAR; the Irish debt-crisis model, which has already seen a banking crisis (the GFC), loses the least; the COVID model, facing an exogenous shock unlike anything in its history, falls in between.	Frozen vs annually retrained models; QLIKE ratio per crisis
RQ4 Robustness	Are the SHAP fingerprints stable across random training seeds?	Group-share rankings agree across 5 seeds (mean Spearman ≥ 0.8 , criterion fixed in advance).	Spearman / Kendall rank correlation

5.2 Data

Series	Yahoo ticker	Channel (SHAP group)	Input(s) per day	Role
ISEQ Overall	^ISEQ	domestic	log return; log squared return	Target + own history
S&P 500	^GSPC	us	log return; squared return	US channel (GFC origin)
FTSE 100	^FTSE	uk	log return; squared return	Robustness only: replaces DAX in one run; Brexit check
DAX	^GDAXI	euro	log return; squared return	Euro-area channel (Irish debt crisis)

All raw downloads are cached to data/raw/ the first time, so the study stays reproducible if Yahoo changes.

5.3 Evaluation windows (defined ex post, never model inputs)

Window	Start → end (event)	Origin type	Days	Max drawdown	Median / peak vol
Global Financial Crisis	9 Aug 2007 (BNP Paribas freeze) → 9 Mar 2009 (S&P 500 trough)	Credit/banking, US-led; Irish banking collapse in parallel	402	−78.0%	34.2% / 121.3%
Irish sovereign debt crisis	23 Apr 2010 (Greece requests aid) → 26 Jul 2012 (Draghi); Irish bailout Nov 2010	Sovereign–bank loop, euro area / Ireland	574	−32.3%	18.2% / 72.2%
COVID-19	19 Feb 2020 (S&P 500 peak) → 30 Jun 2020	Exogenous global shock	92	−39.8%	35.9% / 110.5%
Validation: Brexit	1 Jun 2016 → 31 Jul 2016 (referendum 23 Jun)	UK political shock	42	−18.2%	24.9% / 97.7%
Validation: war / rate shock	10 Feb 2022 → 31 Oct 2022 (invasion 24 Feb; ECB hikes from Jul)	European energy + interest-rate shock	185	−27.5%	23.3% / 66.5%
Calm (all other days)	—	—	—	—	median 13.0%

Figures from the pipeline's `data_quality_report.txt`. Volatility = annualised forward 5-day realised volatility. Confirm event dates against a primary source when you write Chapter 2.

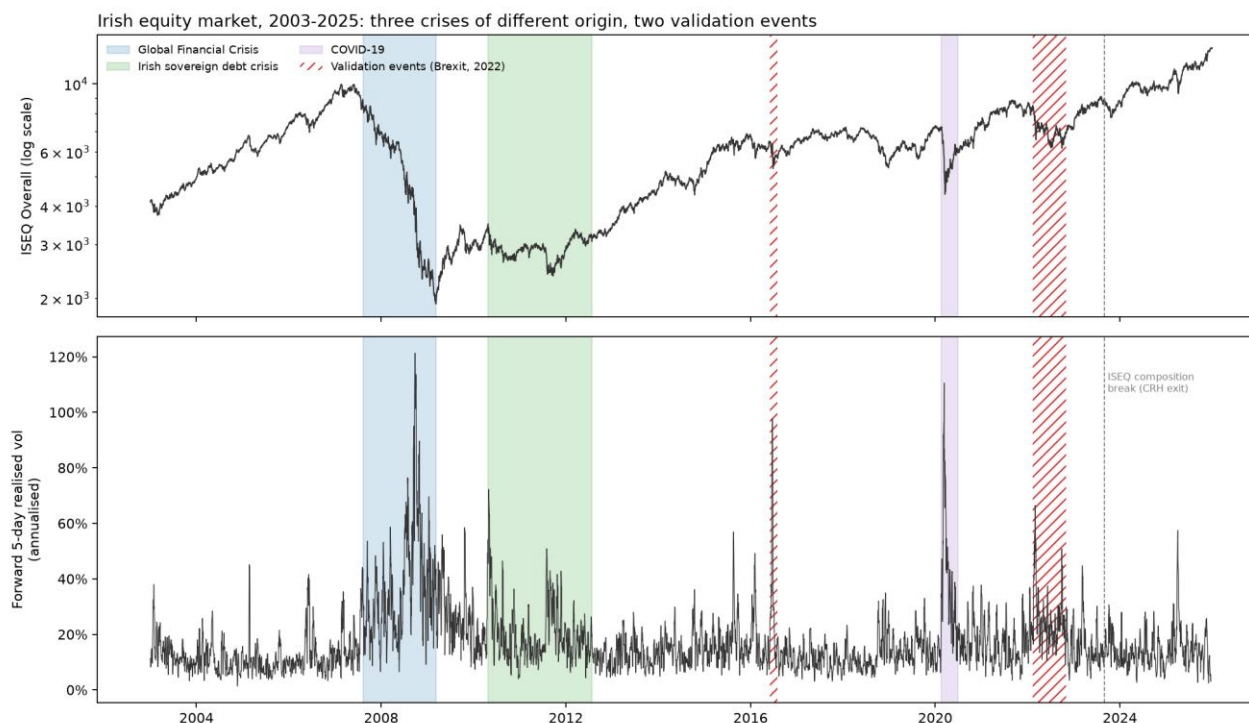


Figure: ISEQ level (top) and forward 5-day realised volatility (bottom), with evaluation windows shaded and the 2023 composition break marked; hatched = validation events. Generated by `code/01_data_pipeline.py`.

5.4 Target variable

- Daily log return $r(t) = \ln(\text{close } t / \text{close } t-1)$.
- Target at forecast origin t : $RV5(t) = r(t+1)^2 + \dots + r(t+5)^2$, i.e. realised variance over the next trading week.
- Models learn log RV5 (more symmetric, always positive). When converting back, apply a bias adjustment estimated on the validation period, because $\exp(\text{average of logs})$ underestimates the average.
- Why weekly, not daily: a single squared return is a very noisy proxy; summing five reduces noise and matches the HAR model's weekly component.
- Robustness: repeat headline results with Parkinson range volatility for 2007–2023.

5.5 Information set and time zones

Forecast issued at 08:00 Dublin time on day $t+1$, before the Irish market opens. By then the Irish close (16:30), London and Frankfurt closes, and the New York close (21:00 Dublin time) of day t are all known. Every input is joined 'as-of' date t or earlier, never $t+1$. When a foreign market was closed on day t , its last known value is carried forward (0.9–2.7% of Irish days). State this explicitly in the Methodology chapter. Cite Hamao, Masulis & Ng (1990) for why trading-hour differences matter.

5.6 Models

Model	Specification	Estimation and refitting	Library
GARCH(1,1)	Daily ISEQ returns ($\times 100$), constant mean, Student-t errors. 5-day forecast = sum of the 1- to 5-step variance forecasts	Refit once a year on an expanding window; parameters fixed within the year while variance updates daily	arch
HAR (log)	log RV5 on log daily, weekly (5-day) and monthly (22-day) average squared returns	OLS, refit annually on the expanding window	statsmodels
LSTM	Look-back 22 days (also try 66); 6 inputs per	Adam (learning rate $1e-3$), MSE on log target,	PyTorch

	day; 1–2 layers of 32 units; dropout 0.2; linear output = log RV5 → tuned result (frozen): look-back 22, 64 units, 2 layers	early stopping on the last 12 months of each training window; refit annually; 5 seeds	
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5.7 Training and validation protocol

- **Expanding window:** first training period Jan 2003 – Dec 2006 (4 years); test 2007; add 2007 to training; test 2008; and so on to Aug 2023 (17 test years).
- **Validation:** the last 12 months of each training window, used only for early stopping and bias adjustment.
- **Scaling:** each scaler is fitted on its own training window only, then applied to the next test year.
- **Seeds:** 5 random seeds per refit; report the mean forecast and use the spread for RQ4.
- **Tuning:** small, fixed grid (look-back 22/66; units 32/64), chosen on validation loss in the first windows only, then frozen. Say so; do not tune on test data.
- **RQ3 frozen models:** train to 8 Aug 2007 then test on the GFC; to 22 Apr 2010 then test on the euro crisis; to 18 Feb 2020 then test on COVID, with no refitting. Compare with the annually refitted model.

5.8 Evaluation

- **Losses:** MSE on the variance level and QLIKE = $RV/F - \ln(RV/F) - 1$ (Patton, 2011). QLIKE is primary.
- **Significance:** Diebold–Mariano with a Newey–West variance (4 lags, because 5-day forecasts overlap) and the Harvey–Leybourne–Newbold small-sample correction.
- **By window:** report losses for each crisis and calm periods. Short windows (COVID, 92 days) have little statistical power; describe them, do not over-test them.
- **Optional:** Mincer–Zarnowitz regression for forecast bias; Model Confidence Set (Hansen, Lunde & Nason, 2011) only if time allows.

5.9 Explainability

- **Method:** Captum GradientShap (an expected-gradients approximation to SHAP) on the trained PyTorch LSTM. Integrated Gradients as a cross-check.
- **Background (baseline) set:** 200 randomly drawn input windows from the same training period.
- **Days explained (as run):** every day in each window plus every 10th calm day — 1,547 days × 5 seeds (decision 25 Sep 2026; full windows allow a block bootstrap over consecutive days).
- **Aggregation:** sum absolute attributions over the 22 look-back days and over the inputs in each channel, then convert to shares of the total. Three channels: domestic, us, euro (uk in the robustness run).
- **Uncertainty:** block-bootstrap the sampled days to put confidence intervals on each share.
- **Sanity check:** grouped permutation importance should rank channels broadly the same way. (Not run so far — optional; the exploratory Diebold–Yilmaz comparison, E1, now plays a similar cross-checking role.)
- **Interpretation rule:** attributions describe what drives the model’s forecasts, not causes. Call them ‘fingerprints consistent with transmission’, never proof of contagion.

5.10 Scope

Tier	Work
MUST	Data pipeline (done) · GARCH, HAR, LSTM · walk-forward forecasts · MSE/QLIKE + DM-HLN · grouped SHAP by window · RQ3 frozen-model test
SHOULD	SHAP stability across 5 seeds (RQ4) · 2022 validation event · FTSE-for-DAX robustness run (incl. Brexit) · Parkinson robustness · post-2023 robustness
EXPLORATORY (A1)	E1 Diebold–Yilmaz spillovers vs SHAP · E2 volatility paradox — done 30 Sep 2026, reported separately from the hypotheses
DEFER (PhD)	Monte Carlo Dropout vs GARCH simulation intervals · TVP-VAR connectedness · Irish bank stocks · sovereign bonds · more countries (Diebold–Yilmaz moved to EXPLORATORY)

5.11 Methodological framing (for the Methodology chapter)

- Philosophy: positivist; approach: deductive (hypotheses stated first); design: quantitative, longitudinal secondary-data study (Saunders, Lewis & Thornhill, 2019). Note: unlike the old proposal, this is not cross-sectional.
- Process model: CRISP-DM, which DBS examiners know. Map each phase to a section.
- Artefact: a reproducible, open-data forecasting-and-explanation pipeline (code, cached data, figures). Evaluation language for an Information Systems audience: Hevner et al. (2004).
- Ethics: public market data only; no personal data; no human participants.

5.12 Novelty statement (narrow and defensible)

Prior work has forecast volatility with LSTMs (Bucci, 2020), explained volatility models with SHAP (Ahmed, 2026, preprint), and measured crisis spillovers with econometrics and deep learning (Sahiner, 2023; Corbet & Twomey, 2015). This study contributes by (1) using grouped SHAP attributions as falsifiable ‘fingerprints’ compared across three crises of different origin, checked against two single-origin validation events, (2) testing whether models trained on earlier crises generalise to the next, and (3) doing so for a small open economy exposed to every one of those shocks. Plan the claim at this level. Do not claim to be ‘the first to use SHAP for volatility’.

5.13 Compute budget (estimate, confirm in week 3)

About 17 annual refits $\times$ 5 seeds = ~85 small LSTM fits on up to ~5,000 samples each. At an estimated 1–3 minutes per fit on a laptop CPU, that is 1.5–4 hours in total, which can be run overnight or on free Google Colab.

GARCH/HAR: seconds. GradientShap on ~1,200 sampled days: minutes. Time one fit in week 3 and update this.

Actual (25–30 Sep 2026, 12 CPU threads): tuning + 85 LSTM fits $\approx$ 15 min; SHAP $\approx$ 21 min (2 workers, 7.7 GB RAM); benchmarks $\approx$ 10 s; E1 and E2 a few seconds each. University GPU PCs are available if needed.

5.14 Exploratory extensions (Amendment A1, 30 Sep 2026)

Chosen after the main results were known, so they are exploratory: no pass/fail rule, no effect on any hypothesis verdict, reported in full. The specification was committed to Git (PREREGISTRATION.md $\rightarrow$ Amendments, commit 9c89f79) before either script was run.

Extension	Question	Method and data	Result
E1 Spillovers vs SHAP (code/11_connectedness.py)	Does an econometric spillover measure tell the same story as the SHAP fingerprints?	Diebold & Yilmaz (2012): VAR(4) on log 5-day RV of ISEQ, S&P 500, DAX; 200-day rolling; generalized FEVD at 10 days; ISEQ row averaged over the SHAP days. Sensitivity: daily r^2 , 100/300-day windows	Weak agreement: 4 of 8 direction checks, 2 of 5 rankings (same in all sensitivity runs). Both show the US highest in COVID-19; in 2022 the spillover euro share rises (0.381) while SHAP’s falls. Total spillover 34.0%; ISEQ net receiver (–8.4)
E2 Volatility paradox (code/12_volatility_paradox.py)	Were the crises preceded by unusually calm markets?	Danielsson et al. (2018): FRED/OECD monthly prices (Ireland from 1955); annual volatility July–June; one-sided HP trend $\lambda = 5,000$; mean below-trend part over 5 pre-crisis years; percentile in own history	Before the GFC only Germany was unusually calm (4th percentile); Ireland 40th, US 42nd. No market unusually calm before COVID-19

6. Risk register

Risk	Likelihood	Impact	Mitigation
LSTM does not beat HAR/GARCH	Medium–high	Low	Pre-registered; a well-argued negative result is publishable. RQ2–RQ4 still stand

SHAP fingerprints unstable across seeds	Medium	High	RQ4 tests it; average over seeds; report honestly
SHAP tooling breaks	Medium	High	PyTorch + Captum; toy test in week 2; Integrated Gradients fallback
Correlated inputs blur channels	Medium	Medium	Only three markets; FTSE kept out of the main model (0.84 overlap with DAX); discuss in Limitations
Yahoo changes or removes data	Low	High	Raw data cached on first download; keep a backup copy
Overclaiming causality	Medium	High	Fixed interpretation rule (5.9); supervisor review of Discussion
Time squeeze	High	High	MUST tier first; results by early November; SHOULD items only after the core is done
Integrity concerns about AI use	Low if managed	Very high	Section 11 protocol; declare everything
Validation events contradict hypotheses	Medium	Medium	Pre-registered; report honestly; it is evidence about SHAP's limits
Extra analyses after the results look like p-hacking	Medium	High	Dated amendment committed before running; labelled exploratory; no pass/fail; reported in full; never used to change a verdict

7. Pathway

7.1 Dissertation — 12-week plan (adjust to the official DBS deadline)

Week	Dates (approx.)	Build	Learn / read	Deliverable
0	now – 4 Oct	Pipeline + chart (done)	Reinhart & Rogoff ch.1; Corsi (2009)	1-page pitch + chart for supervisor
1	5–11 Oct	Supervisor sign-off; Git repo; PREREGISTRATION.md	Forbes & Rigobon; Hansen & Lunde	Approved topic; hypotheses committed
2	12–18 Oct	Install arch, statsmodels, PyTorch, Captum; toy LSTM + GradientShap test	Lundberg & Lee; Captum docs	SHAP proven to work
3	19–25 Oct	GARCH + HAR walk-forward; loss functions; time one LSTM fit	Patton (2011); DM + HLN	Benchmark forecasts
4–5	26 Oct – 8 Nov	LSTM walk-forward, 5 seeds	Bucci (2020); Christensen et al. (2023)	First LSTM vs benchmarks table; first draft of Lit Review 2.2–2.4
6	9–15 Nov	DM-HLN tests overall and by window; RQ3 frozen models	Sahiner (2023)	RQ1 + RQ3 results
7	16–22 Nov	Grouped GradientShap by window; bootstrap intervals	Aas et al. (2021); Kumar et al. (2020)	RQ2 results
8	23–29 Nov	Seed stability; 2022 check; FTSE robustness run (Brexit); Parkinson robustness	Corbet & Twomey (2015)	RQ4 + robustness
9	30 Nov – 6 Dec	Write Methodology + Results	—	Draft ch.3–4 to supervisor
10	7–13 Dec	Polish Literature Review (first draft done in week 5); write Discussion	Crisis history sources	Final ch.2, draft ch.5
11	14–20 Dec	Introduction, Conclusion, Abstract; references	—	Full draft
12	by deadline	Proofread; reproducibility check; AI declaration	—	Submission

Status at 30 Sep 2026: the build work planned for weeks 1–8 is done except the FTSE, look-back-66 and post-2023 robustness runs, and two exploratory extensions have been added (5.14). The remaining weeks are for reading, the robustness runs and writing.

Future work beyond the MSc: extend the exploratory Diebold–Yilmaz comparison (E1) to time-varying (TVP-VAR) connectedness, the approach used by Irish finance groups; add honest uncertainty (Monte Carlo Dropout vs GARCH simulation); test the volatility paradox properly on a panel; extend to Irish banks, sovereign bonds and more small open economies; build an explainable early-warning system for crises.

8. Reading list

How to read: pass 1 (10 minutes) — abstract, introduction, figures, conclusion. Pass 2 (1 hour) — method and results, noting data, model, test and finding. Pass 3 only for the six ★papers. Keep one note per paper in Zotero (free) and use Harvard style as DBS requires.

Where to get them: DBS library (ScienceDirect, JSTOR, Oxford Academic), Google Scholar ‘All versions’ for free PDFs, SSRN, arXiv, and authors’ web pages (e.g. Patton’s paper is free on his Duke page).

✓= details confirmed in this session. Others are standard references believed correct: confirm volume and pages in the library before citing.

8.1 Start here (weeks 0–2)

	Reference	Why read it	✓
★	Reinhart, C. & Rogoff, K. (2009) This Time Is Different: Eight Centuries of Financial Folly. Princeton UP.	Your title and historical frame; crisis typology	–
★	Corsi, F. (2009) A simple approximate long-memory model of realized volatility. J. Financial Econometrics 7(2), 174–196.	The HAR benchmark	–
★	Bucci, A. (2020) Realized volatility forecasting with neural networks. J. Financial Econometrics 18(3), 502–531.	Closest LSTM-volatility evidence	✓
★	Christensen, K., Siggaard, M. & Veliyev, B. (2023) A machine learning approach to volatility forecasting. J. Financial Econometrics 21(5), 1680–1727.	ML vs HAR; how to set up the comparison	✓
★	Lundberg, S. & Lee, S.-I. (2017) A unified approach to interpreting model predictions. NeurIPS 30.	SHAP itself	–
★	Forbes, K. & Rigobon, R. (2002) No contagion, only interdependence. J. Finance 57(5), 2223–2261.	Why rising co-movement in crises is not automatically contagion	–
	Corbet, S. & Twomey, C. (2015) European equity market contagion: an empirical application to Ireland’s sovereign debt crisis. European Financial and Accounting Journal 2015(3), 15–34.	Twomey’s Irish evidence; compare with your euro-crisis window	✓
	Sahiner, M. (2023) Volatility spillovers and contagion during major crises: an early warning approach based on a deep learning model. Computational Economics.	Closest prior work: cite and differentiate	✓

8.2 Volatility econometrics and forecast evaluation

	Reference	Why read it	✓
	Engle, R. (1982) Autoregressive conditional heteroscedasticity... Econometrica 50(4), 987–1007.	ARCH origin	–
	Bollerslev, T. (1986) Generalized autoregressive conditional heteroskedasticity. J. Econometrics 31(3), 307–327.	GARCH	–
	Hansen, P. & Lunde, A. (2005) A forecast comparison of volatility models: does anything beat a GARCH(1,1)? J. Applied Econometrics 20(7), 873–889.	Why GARCH is a tough benchmark	–
	Andersen, T. & Bollerslev, T. (1998) Answering the skeptics... International Economic Review 39(4), 885–905.	Noisy proxies make good models look bad	–
	Andersen, T., Bollerslev, T., Diebold, F. & Labys, P. (2003) Modeling and forecasting realized volatility. Econometrica 71(2), 579–625.	Realised variance	–
	Patton, A. (2011) Volatility forecast comparison using imperfect volatility proxies. J. Econometrics 160(1), 246–256.	Why QLIKE	✓
	Diebold, F. & Mariano, R. (1995) Comparing predictive accuracy. J. Business & Economic Statistics 13(3), 253–263.	DM test	–
	Harvey, D., Leybourne, S. & Newbold, P. (1997) Testing the equality of prediction mean squared errors. International J. Forecasting 13(2), 281–291.	HLN correction	–
	Parkinson, M. (1980) The extreme value method for estimating the variance of the rate of return. J. Business 53(1), 61–65.	Robustness estimator	–
	Garman, M. & Klass, M. (1980) On the estimation of security price volatilities from historical data. J. Business 53(1), 67–78.	Explains why you rejected it	–
	Cont, R. (2001) Empirical properties of asset returns: stylized facts and statistical issues. Quantitative Finance 1(2), 223–236.	Volatility clustering, fat tails	–

8.3 Deep learning and explainability

	Reference	Why read it	✓
	Hochreiter, S. & Schmidhuber, J. (1997) Long short-term memory. <i>Neural Computation</i> 9(8), 1735–1780.	LSTM	–
	Srivastava, N. et al. (2014) Dropout. <i>J. Machine Learning Research</i> 15, 1929–1958.	Regularisation	–
	Fischer, T. & Krauss, C. (2018) Deep learning with LSTM networks for financial market predictions. <i>European J. Operational Research</i> 270(2), 654–669.	LSTM in finance	–
	Kim, H. & Won, C. (2018) Forecasting the volatility of stock price index: a hybrid model integrating LSTM with multiple GARCH-type models. <i>Expert Systems with Applications</i> 103, 25–37.	LSTM + GARCH hybrids	–
	Sundararajan, M., Taly, A. & Yan, Q. (2017) Axiomatic attribution for deep networks. <i>ICML</i> .	Integrated Gradients	–
	Erion, G. et al. (2021) Improving performance of deep learning models with axiomatic attribution priors and expected gradients. <i>Nature Machine Intelligence</i> 3, 620–631.	Expected gradients = GradientSHAP	–
	Aas, K., Jullum, M. & Løland, A. (2021) Explaining individual predictions when features are dependent. <i>Artificial Intelligence</i> 298, 103502.	Correlated features	–
	Kumar, I. et al. (2020) Problems with Shapley-value-based explanations as feature importance measures. <i>ICML</i> .	Limits of SHAP	–
	Rudin, C. (2019) Stop explaining black box machine learning models for high stakes decisions... <i>Nature Machine Intelligence</i> 1, 206–215.	The counter-argument to discuss	–
	Ahmed, A. (2026) Explainable deep learning for financial volatility forecasting: an LSTM-Attention-SHAP framework. SSRN preprint.	Recent close work; preprint, not peer-reviewed	✓
	Bergmeir, C. & Benítez, J. (2012) On the use of cross-validation for time series predictor evaluation. <i>Information Sciences</i> 191, 192–213.	Time-series validation	–
	Feng, L., Shi, J. & Kutan, A. (2026) Your fear is (partly) mine: the role of non-VIX volatility in forecasting regional stock market volatility using interpretable machine learning. <i>J. International Money and Finance</i> 160, 103467.	Closest new work: interpretable ML for cross-market volatility — must cite	✓
	Brini, A. (2026) Forecasting realized volatility with time series foundation models: a comparison with econometric benchmarks. arXiv 2607.05291 (preprint).	Nine foundation models vs HAR on 50 assets: no general advantage — supports your H1 result	✓

8.4 Contagion and transmission

	Reference	Why read it	✓
	King, M. & Wadhwani, S. (1990) Transmission of volatility between stock markets. <i>Review of Financial Studies</i> 3(1), 5–33.	Classic transmission paper	–
	Hamao, Y., Masulis, R. & Ng, V. (1990) Correlations in price changes and volatility across international stock markets. <i>Review of Financial Studies</i> 3(2), 281–307.	Time zones and spillovers	✓
	Engle, R., Ito, T. & Lin, W. (1990) Meteor showers or heat waves? <i>Econometrica</i> 58(3), 525–542.	Domestic vs foreign volatility	–
	Bekaert, G., Harvey, C. & Ng, A. (2005) Market integration and contagion. <i>J. Business</i> 78(1), 39–69.	Integration vs contagion	–
	Pericoli, M. & Sbracia, M. (2003) A primer on financial contagion. <i>J. Economic Surveys</i> 17(4), 571–608.	Definitions of contagion	–
	Dungey, M. et al. (2005) Empirical modelling of contagion: a review of methodologies. <i>Quantitative Finance</i> 5(1), 9–24.	Method overview	–
	Diebold, F. & Yilmaz, K. (2009) <i>Economic Journal</i> 119(534), 158–171; (2012) <i>International J. Forecasting</i> 28(1), 57–66.	Spillover index — the method of exploratory E1 (2012 set-up checked in the full text)	✓
	Bekaert, G., Ehrmann, M., Fratzscher, M. & Mehl, A. (2014) The global crisis and equity market contagion. <i>J. Finance</i> 69(6), 2597–2649.	GFC contagion: small from the US, large from domestic markets with weak fundamentals	✓
	Kearney, C. (2000) The determination and international transmission of stock market volatility. <i>Global Finance Journal</i> 11, 31–52.	Volatility transmission (Irish author)	✓
	Pesaran, M.H. & Shin, Y. (1998) Generalized impulse response analysis in linear multivariate models. <i>Economics Letters</i> 58(1), 17–29.	Order-invariant (generalized) variance decomposition used in E1	–
	Antonakakis, N., Chatziantoniou, I. & Gabauer, D. (2020) Refined measures of dynamic connectedness based on time-varying parameter vector autoregressions. <i>J. Risk and Financial Management</i> 13(4), 84.	TVP-VAR connectedness — the PhD extension of E1	–

8.5 Crisis history (for Chapter 2 and your Discussion)

Reference	Why read it	✓
Kindleberger, C. & Aliber, R. (2015) Manias, Panics and Crashes. 7th edn. Palgrave Macmillan.	Crisis anatomy	–
Minsky, H. (1986) Stabilizing an Unstable Economy. Yale UP; Minsky (1992) The Financial Instability Hypothesis. Levy Institute Working Paper 74.	Stability breeds instability	–
Fama, E. (1970) Efficient capital markets: a review of theory and empirical work. J. Finance 25(2), 383–417.	Market efficiency: the counterpoint to Minsky	–
Brunnermeier, M. (2009) Deciphering the liquidity and credit crunch 2007–2008. J. Economic Perspectives 23(1), 77–100.	GFC mechanics	–
Lane, P. (2012) The European sovereign debt crisis. J. Economic Perspectives 26(3), 49–68.	Euro crisis (by the later Central Bank of Ireland governor)	–
Baker, S. et al. (2020) The unprecedented stock market reaction to COVID-19. Review of Asset Pricing Studies 10(4), 742–758.	COVID shock	–
Akhtaruzzaman, M., Boubaker, S. & Sensoy, A. (2021) Financial contagion during COVID-19 crisis. Finance Research Letters 38, 101604.	COVID contagion via financial firms	✓
Li, H. (2020) Volatility spillovers across European stock markets under the uncertainty of Brexit. Economic Modelling 84, 1–12.	Brexit: UK influence on European markets fell before the vote; co-volatility persisted after it (competing evidence)	✓
Boungou, W. & Yatié, A. (2022) The impact of the Ukraine–Russia war on world stock market returns. Economics Letters 215, 110516.	2022 validation event	✓
Tooze, A. (2018) Crashed: How a Decade of Financial Crises Changed the World. Allen Lane.	Narrative history of 2008–2018 (you will enjoy it)	–
Galbraith, J.K. (1955) The Great Crash 1929.	Historical context only	–
Danielsson, J., Valenzuela, M. & Zer, I. (2018) Learning from history: volatility and financial crises. Review of Financial Studies 31(7), 2774–2805.	The volatility paradox: prolonged low volatility precedes banking crises — basis of E2	✓

8.6 Ireland

Reference	Why read it	✓
Honohan, P. (2010) The Irish Banking Crisis: Regulatory and Financial Stability Policy 2003–2008. Central Bank of Ireland.	Official account	–
Regling, K. & Watson, M. (2010) A Preliminary Report on the Sources of Ireland's Banking Crisis.	Home-grown vs imported	–
Nyberg, P. (2011) Misjudging Risk: Causes of the Systemic Banking Crisis in Ireland.	Official inquiry	–
Kelly, M. (2009) The Irish credit bubble. UCD Centre for Economic Research WP09/32.	The domestic credit bubble behind the GFC in Ireland	✓
Whelan, K. (2014) Ireland's economic crisis: the good, the bad and the ugly. J. Macroeconomics 39(B), 424–440.	Overview of the Irish crisis	✓
Acharya, V., Drechsler, I. & Schnabl, P. (2014) A Pyrrhic victory? Bank bailouts and sovereign credit risk. J. Finance 69(6), 2689–2739.	The bank–sovereign loop (Ireland's 2008 guarantee)	✓
Kearney, C. (1998) The causes of volatility in a small, internationally integrated stock market: Ireland, July 1975–June 1994. J. Financial Research 21(1).	Irish market volatility; small open economy	✓ partial
Gallagher, L. & Twomey, C. (1998) Identifying the source of mean and volatility spillovers in Irish equities: a multivariate GARCH analysis. The Economic and Social Review 29(4), 341–356.	UK → Irish spillovers	✓
Irish Times (2023–2024) reporting on CRH, Flutter and Smurfit Kappa leaving the ISEQ.	Justifies the Aug 2023 sample end	✓

8.7 Methods, textbooks and free resources

Reference	Why read it	✓
Brooks, C. (2019) Introductory Econometrics for Finance. 4th edn. Cambridge UP.	GARCH, VAR, testing — your main econometrics text	–
Tsay, R. (2010) Analysis of Financial Time Series. 3rd edn. Wiley.	Volatility models in depth	–

	Hyndman, R. & Athanasopoulos, G. Forecasting: Principles and Practice (3rd edn, free at otexts.com/fpp3).	Forecast evaluation, time-series cross-validation	–
	Zhang, A. et al. Dive into Deep Learning (free at d2l.ai), RNN/LSTM chapters.	LSTM in PyTorch	–
	Molnar, C. Interpretable Machine Learning (free online).	SHAP explained clearly	–
	López de Prado, M. (2018) Advances in Financial Machine Learning. Wiley.	Leakage and backtesting pitfalls (chapters 7 and 11)	–
	Saunders, M., Lewis, P. & Thornhill, A. (2019) Research Methods for Business Students. 8th edn. Pearson.	Methodology chapter	–
	Hevner, A. et al. (2004) Design science in information systems research. MIS Quarterly 28(1), 75–105.	Artefact (design-science) framing	–
	Hodrick, R. & Prescott, E. (1997) Postwar U.S. business cycles: an empirical investigation. J. Money, Credit and Banking 29(1), 1–16.	The HP filter used in E2	–

9. What to learn — finance and econometrics

Concept	What it is	Why you need it	Learn from	Week
Log returns	$\ln(P_t/P_{t-1})$; additive over time	Every input and the target	Brooks ch.2	0
Variance, volatility, annualising	$\text{Vol} = \sqrt{\text{variance}}$; $\times \sqrt{252}$ for yearly	Reading and reporting results	Brooks; Christoffersen	0
Stylised facts	Fat tails, volatility clustering, leverage effect	Why volatility is forecastable	Cont (2001)	1
Realised variance and proxies	Sum of squared returns; range estimators	Your target and its noise	ABDL (2003); Patton (2011)	1
ARCH/GARCH(1,1)	Variance = $\omega + \alpha \cdot \text{shock}^2 + \beta \cdot \text{last variance}$; persistence $\alpha + \beta$	Benchmark 1	Brooks ch.9; Engle; Bollerslev	2–3
Student-t errors	Fat-tailed error distribution for GARCH	Better GARCH fit	Brooks ch.9	3
Multi-step variance forecasts	Iterating GARCH forward h days	5-day target	arch docs	3
HAR model	Daily, weekly, monthly volatility components in one regression	Benchmark 2	Corsi (2009)	3
Stationarity; ADF test	Stable mean/variance over time	Justify log RV modelling	Brooks ch.8	2
Efficient Market Hypothesis	Prices reflect available information	Why returns are hard but volatility easier to forecast	Fama (1970)	1
Contagion vs interdependence	Crisis-only increase in linkages vs normal linkages	Framing RQ2 honestly	Forbes & Rigobon; Pericoli & Sbracia	1–2
Spillovers and connectedness	Share of one market's forecast error variance caused by another	Exploratory E1; literature review	Diebold & Yilmaz	now
Minsky's instability hypothesis	Hedge $\rightarrow$ speculative $\rightarrow$ Ponzi finance	Theory for bubbles and the GFC	Minsky (1992)	2
Sovereign–bank 'doom loop'	Weak banks and weak governments drag each other down	Euro/Irish crisis mechanism	Lane (2012); Honohan (2010)	4
Small open economy	Economy dominated by external shocks	Why Ireland is a good test case	Lane (2012)	2
VIX	Implied 30-day S&P 500 volatility; 'fear gauge'	Context only (not a model input)	CBOE material	2
Asynchronous trading / time zones	Markets close at different times	No look-ahead	Hamao et al. (1990)	1
Vector autoregression (VAR)	Each market regressed on the recent past of all markets	E1	Brooks ch.7	now
Generalized variance decomposition	Order-invariant split of forecast uncertainty by source market	E1 (Diebold–Yilmaz, 2012)	Pesaran & Shin (1998); Diebold & Yilmaz (2012)	now
Hodrick–Prescott filter (one-sided)	Trend vs cycle, estimated from past data only	E2	Hodrick & Prescott (1997); Danielsson et al. (2018)	now
Volatility paradox	Long calm periods encourage risk-taking; low volatility precedes crises	E2; Minsky discussion	Danielsson et al. (2018)	now

10. What to learn — machine learning, explainability, statistics and tools

10.1 Machine learning

Concept	What it is	Why you need it	Learn from	Week
Look-ahead bias and leakage	Using information not available at forecast time	The most common examiner criticism	López de Prado (2018); FPP3	1
Walk-forward / expanding window	Train on the past, test on the next period, repeat	Your whole evaluation design	FPP3 ch.5; Bergmeir & Benítez	2
Train/validation/test in time series	Validation = last part of training, never the future	Early stopping without leakage	FPP3	2
Scaling fitted on training only	Scaler learns mean/SD from training data	Avoid leakage	scikit-learn docs	2
RNNs and vanishing gradients	Why plain RNNs forget long history	Justifies LSTM	d2l.ai ch.9	2
LSTM gates	Forget, input, output gates and cell state	Explain your model in ch.3	Hochreiter & Schmidhuber; d2l.ai ch.10	2
Sequence windows (look-back)	Input = last L days	Model design; SHAP over time	d2l.ai	3
Loss, optimiser, learning rate	MSE; Adam; step size	Training	d2l.ai	3
Early stopping, dropout, weight decay	Ways to prevent overfitting	Training	Srivastava et al.	3
Random seeds and non-determinism	Same code, different results	RQ4 and reproducibility	PyTorch docs	4
Hyperparameter tuning without test data	Small grid on validation only	Honest results	FPP3	4

10.2 Explainable AI

Concept	What it is	Why you need it	Learn from	Week
Shapley values	Fair split of a payoff among players; efficiency, symmetry, dummy, additivity	Theory behind SHAP	Molnar ch. Shapley	2
SHAP family	KernelSHAP, TreeSHAP, DeepSHAP, GradientSHAP	Justify GradientSHAP	Lundberg & Lee; Molnar	2
Integrated Gradients / expected gradients	Path-based attributions for neural networks	How GradientSHAP works; cross-check	Sundararajan et al.; Erion et al.	7
Baseline (background) distribution	Reference inputs attributions are measured against	Results depend on it; must justify	Captum docs	7
Local vs global explanations	One forecast vs average behaviour	Window-level shares are global	Molnar	7
Grouped attributions	Summing attributions within a channel	Your fingerprint measure	Aas et al.	7
Correlated-feature problem	Credit split arbitrarily among correlated inputs	Limitations; grouping	Aas et al.; Kumar et al.	7
Explanation stability	Do explanations change with seeds/data?	RQ4	Kumar et al.	8
Permutation importance	Accuracy drop when a group is shuffled	Sanity check for SHAP	Molnar	8
Explanation ≠ causation	Attributions describe the model, not the world	Interpretation rule	Rudin (2019)	8

10.3 Statistics and forecast evaluation

Concept	What it is	Why you need it	Learn from	Week
MSE, RMSE, MAE	Standard error measures	Reporting	FPP3	3
QLIKE	Loss robust to noisy volatility proxies	Primary loss	Patton (2011)	3
Diebold–Mariano test	Is one model's average loss significantly lower?	RQ1	Diebold & Mariano	6
HAC / Newey–West variance	Standard errors robust to autocorrelation	Overlapping 5-day forecasts	Brooks	6
Harvey–Leybourne–Newbold correction	Small-sample adjustment to DM	RQ1	HLN (1997)	6

Block bootstrap	Resampling blocks of days to keep time dependence	Confidence intervals for SHAP shares	Brooks; any bootstrap tutorial	7
Spearman / Kendall rank correlation	Agreement between rankings	RQ4	any statistics text	8
Mincer–Zarnowitz regression	Regress outcome on forecast; ideal: intercept 0, slope 1	Optional bias check	Patton (2011)	6
Model Confidence Set	Set of models not significantly worse than the best	Optional	Hansen et al. (2011)	—

10.4 Tools and environment

Tool	Use	Status on your PC
Python 3.13 (py -3.13)	Everything	Installed
pandas, numpy, yfinance, matplotlib	Data and charts	Installed in this session (user-level)
scikit-learn	Scalars, metrics	Installed
arch	GARCH	Installed (8.0.0)
statsmodels	HAR (OLS), ADF, Newey–West	Installed (0.15.0)
PyTorch (CPU) + Captum	LSTM and GradientShap	Installed and tested (torch 2.14.0 CPU, Captum 0.9.0)
Git + GitHub	Version control; timestamped pre-registration	Set up; public repository on GitHub (you push)
Zotero	References, Harvard style	Install
Google Colab	Free extra compute if needed	Optional; university GPU PCs also available

Project folder: Desktop\Dissertation\ → code\01–12 (all run), data\raw (cached Yahoo and FRED downloads), data\processed\dataset.csv, results\ (all tables), figures\ (all charts), docs\, notes\. See the README for the current results.

11. Integrity and AI-use protocol

- Check DBS’s current policy on generative AI before writing. Follow it exactly.
- Declare AI assistance in the Methodology the way DBS requires; over-declare rather than under-declare.
- Write every sentence of the dissertation yourself. Use this plan for decisions, not text.
- Understand every line of code you submit; be able to explain it in a viva or PhD interview.
- Keep a dated decision log (notes/decisions.md) and commit hypotheses before results.
- Do not reuse text, tables or numbers from other dissertations or from AI-generated drafts.
- Report results as they come out, including negative ones. Never adjust hypotheses after seeing results without saying so.

12. Results status (30 September 2026)

Facts only, for planning. Interpretation and all dissertation text are yours. Full tables: README.md and the Project Handbook (Part H).

Item	Result
H1 accuracy	Not supported: QLIKE LSTM 0.463 vs GARCH 0.380 and HAR 0.414; LSTM significantly worse (DM-HLN $p = 0.003$ and 0.044). LSTM best in COVID-19 and 2022
H2 fingerprints	H2a and H2b not supported (partly); H2c supported narrowly; H2e not supported; H2d pending (FTSE run)
H3 learning from history	Not supported as ranked: frozen/HAR GFC 4.26 > Irish 1.33 > COVID 0.74 (GFC largest, as predicted)
H4 stability	Not supported: mean Spearman 0.7 in the GFC and Irish windows
Robustness	Parkinson proxy: GARCH still best; LSTM best in COVID. FTSE, look-back 66, post-2023: to run
E1 (exploratory)	SHAP vs Diebold–Yilmaz: weak agreement (4/8 signs, 2/5 rankings)
E2 (exploratory)	Only Germany unusually calm before the GFC; Ireland 40th percentile

Appendix A. Data-quality report (from the pipeline)

^GSPC S&P 500 first=2002-10-01 rows= 5850 Irish days carried
forward=2.6%
^GDAXI DAX (euro-area core; Euro Stoxx 50 only on Yahoo from 2007) first=2002-10-01 rows= 5903 Irish days carried
forward=0.9%
^FTSE FTSE 100 (robustness only - not a main model input) first=2002-10-01 rows= 5873 Irish days carried
forward=0.9%
Daily-return correlations 2003-2023: ISEQ-US 0.48, ISEQ-euro 0.69, ISEQ-UK 0.71, euro-UK 0.83
Main model inputs: ['ret', 'r2', 'us_ret', 'us_r2', 'euro_ret', 'euro_r2']
Robustness inputs (FTSE replaces DAX): ['ret', 'r2', 'us_ret', 'us_r2', 'uk_ret', 'uk_r2']

Why Garman-Klass is rejected (Yahoo ^ISEQ open is often yesterday's close):

2003-2006: stale opens= 8.3% monthly corr(GK, r^2)=0.54 monthly corr(Parkinson, r^2)=0.67
2007-2012: stale opens=37.7% monthly corr(GK, r^2)=0.97 monthly corr(Parkinson, r^2)=0.99
2013-2019: stale opens=75.4% monthly corr(GK, r^2)=0.96 monthly corr(Parkinson, r^2)=0.98
2020-2025: stale opens=10.0% monthly corr(GK, r^2)=0.99 monthly corr(Parkinson, r^2)=0.99

ISEQ by evaluation window (target = forward 5-day realised variance):

calm periods: median annualised vol 13.0%
Global Financial Crisis days= 402 max drawdown=-78.0% median vol=34.2% peak vol=121.3% days above 2x calm vol=68.4%
Irish sovereign debt crisis days= 574 max drawdown=-32.3% median vol=18.2% peak vol= 72.2% days above 2x calm vol=21.4%
COVID-19 days= 92 max drawdown=-39.8% median vol=35.9% peak vol=110.5% days above 2x calm vol=65.2%
Brexit referendum days= 42 max drawdown=-18.2% median vol=24.9% peak vol= 97.7% days above 2x calm vol=47.6%
Ukraine war / rate shock days= 185 max drawdown=-27.5% median vol=23.3% peak vol= 66.5% days above 2x calm vol=36.8%

Training history before the GFC window: 1166 days (4.6 years), all from the calm 2003-07 boom.

Main sample (Jan 2003 - Aug 2023), complete rows: 5243; zero-return days: 2

Days after ISEQ composition break (2023-09-01): 592 -> robustness only, not in the main sample.

Appendix B. Sources checked online in this session

- Corbet & Twomey (2015), IDEAS/RePEc: ideas.repec.org/a/prg/jnlefa/v2015y2015i3id143p15-34.html
- Bucci (2020): doi.org/10.1093/jjfinec/nbaa008
- Christensen, Siggaard & Veliyev (2023): doi.org/10.1093/jjfinec/nbac020
- Patton (2011): ideas.repec.org/a/eee/econom/v160y2011i1p246-256.html
- Hamao, Masulis & Ng (1990): doi.org/10.1093/rfs/3.2.281
- Sahiner (2023): doi.org/10.1007/s10614-023-10412-4
- Ahmed (2026), SSRN abstract 6301119
- SHAP-LSTM issues: github.com/shap/shap/issues/1193 and [/1677](https://github.com/shap/shap/issues/1677)
- Captum GradientShap: captum.ai/api/gradient_shap.html
- Irish Times on CRH/Flutter exits (2 Mar 2024) and Smurfit Kappa exit (8 Sep 2023, 2 Jul 2024)
- 30 Sep 2026 — Diebold & Yilmaz (2012) full text: sas.upenn.edu/~fdiebold/papers/paper99/DirectionofSpillovers_Mar10.pdf
- 30 Sep 2026 — Danielsson, Valenzuela & Zer (2018) accepted version: eprints.lse.ac.uk/91136
- 30 Sep 2026 — FRED series SPASTT01IEM661N, SPASTT01USM661N, SPASTT01DEM661N: fred.stlouisfed.org
- 30 Sep 2026 — Gallagher & Twomey (1998), ESR 29(4), 341–356 (search result; confirm in the ESR archive)
- 30 Sep 2026 — Feng, Shi & Kutun (2026): ideas.repec.org/a/eee/jimfin/v160y2026ics0261560625002025.html
- 30 Sep 2026 — Brini (2026): arxiv.org/abs/2607.05291